Top Ten Open Source Tools for Web Developers

Web apps load like regular Web pages or websites. Their functionalities include working offline, push notifications, and device hardware access that was traditionally available only to native mobile apps. Web development plays an important role in the Internet ecosystem.

Let us kick off this discussion on the best open source tools for Web developers with a closer look at Git.

Git

Most developers use the Git repository for version control. The Git server helps to manage multiple repositories (when many people work in the same repository but on different tasks), to keep track of changes in the code.

Now, let's understand Git's workflow, branching strategy and commands. When working in a team, there are some very important things that everyone should know about Git. These include the following.

The workflow of Git

Git uses the same centralised pattern as a subversion (SVN) workflow but the developer is empowered by its ease of use, compared to SVN. All developers have their own local copy of the entire project while each one works independently, oblivious of all other changes to the project. Developers can add commits to their own local branch and completely ignore the main development/master branch until convenient; i.e., when a bug is fixed or a feature is complete.

In Figure 1, the third stage is the developer's local branch. The Git Terminal is recommended for use although Git GUI is available. The latter might be easier but the developer community finds the terminal more convenient. In Git Terminal, the command line serves to inform the user and give proper error messages; and when the command fails, it also points out the proper solution.

Commands in Git

To create a working directory, use the following command:

```
ssh user@host git init --bare /path/to/repo.git
```

To clone the central repository, use the command shown below:

```
git clone https://user@host/path/to/repo.git OR git clone
ssh://user@host/path/to/repo.git
```

The following command is used to create a local branch and checkout:

```
git checkout -b <your branch name>
```

How to make a change and commit: Once a repository is cloned in the local machine, then the developer can make changes using the standard Git commit process — to edit, stage and commit. If the developer is not familiar with staging, then the other way to prepare a commit without having to include more changes in the working directory is as follows:

```
git status                # View the state of the repo
git add <some-file with give proper path>      # Stage a file
git commit  -m  "Give Proper message"   # this is the local
commit in your local branch
git push origin <your local branch name>
```

***Pushing a new commit in the central repository*:** Once the local repository has new changes committed, these should be